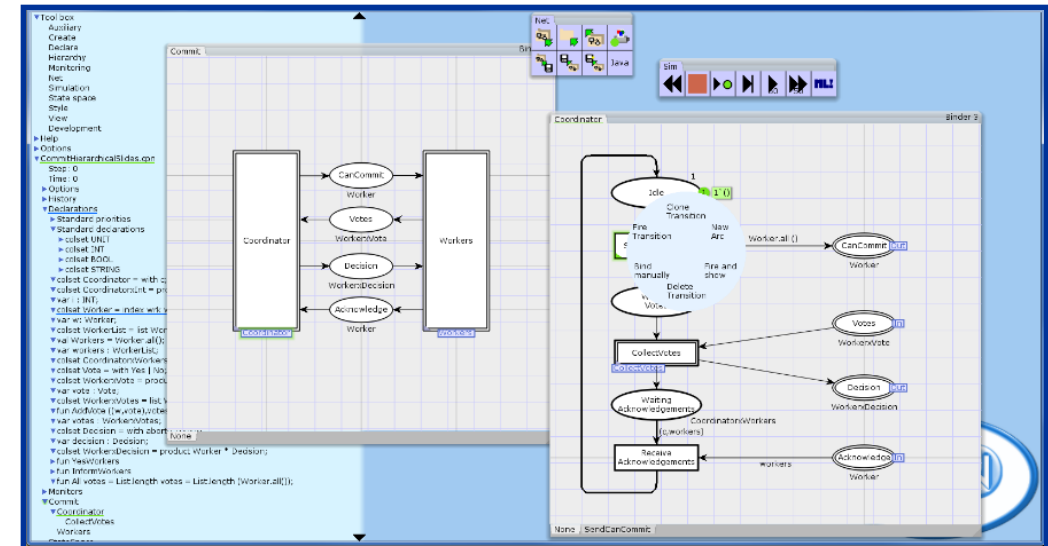
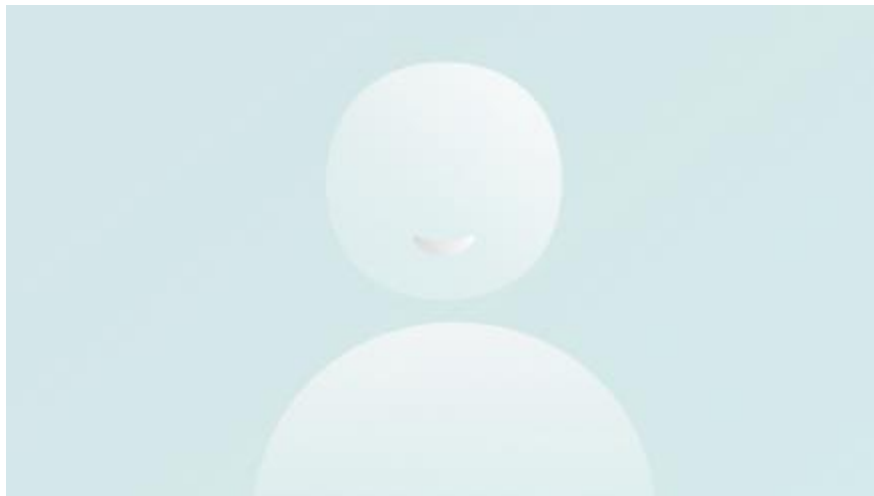


Modelling large systems with Hierarchical Coloured Petri Net Modules



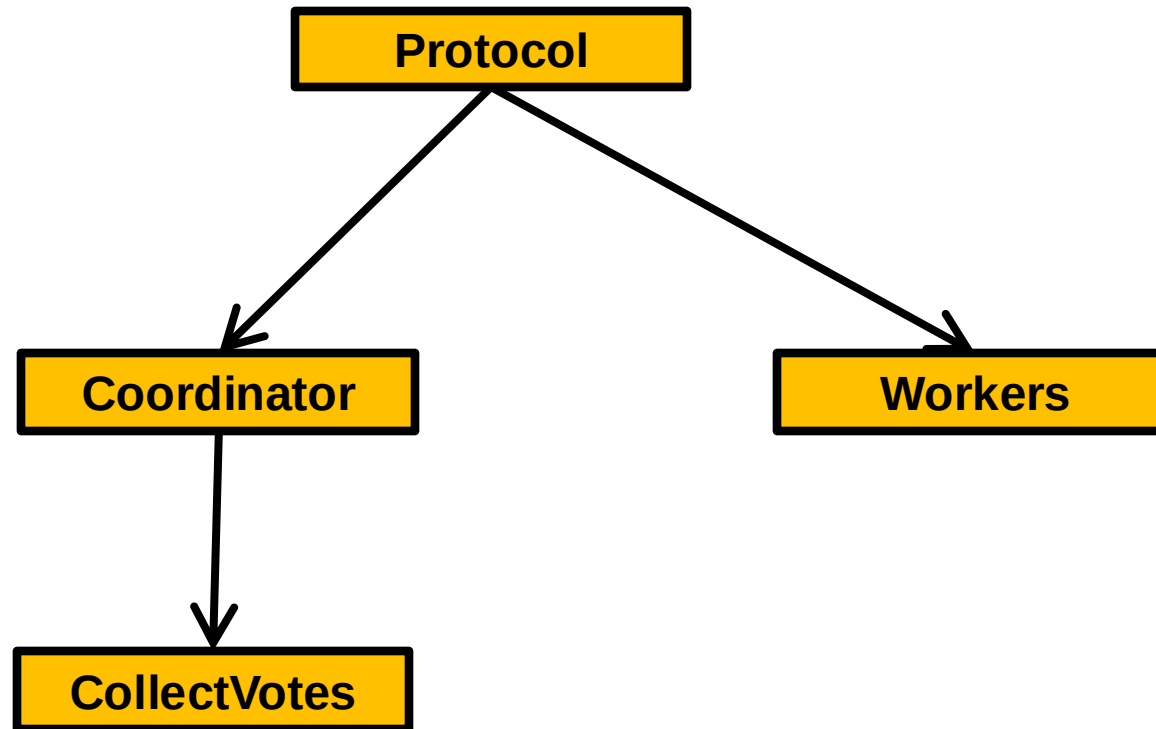
Lars Michael Kristensen
Department of Computer Science, Electrical Engineering, and Mathematical Sciences
Western Norway University of Applied Sciences

Motivation and concepts

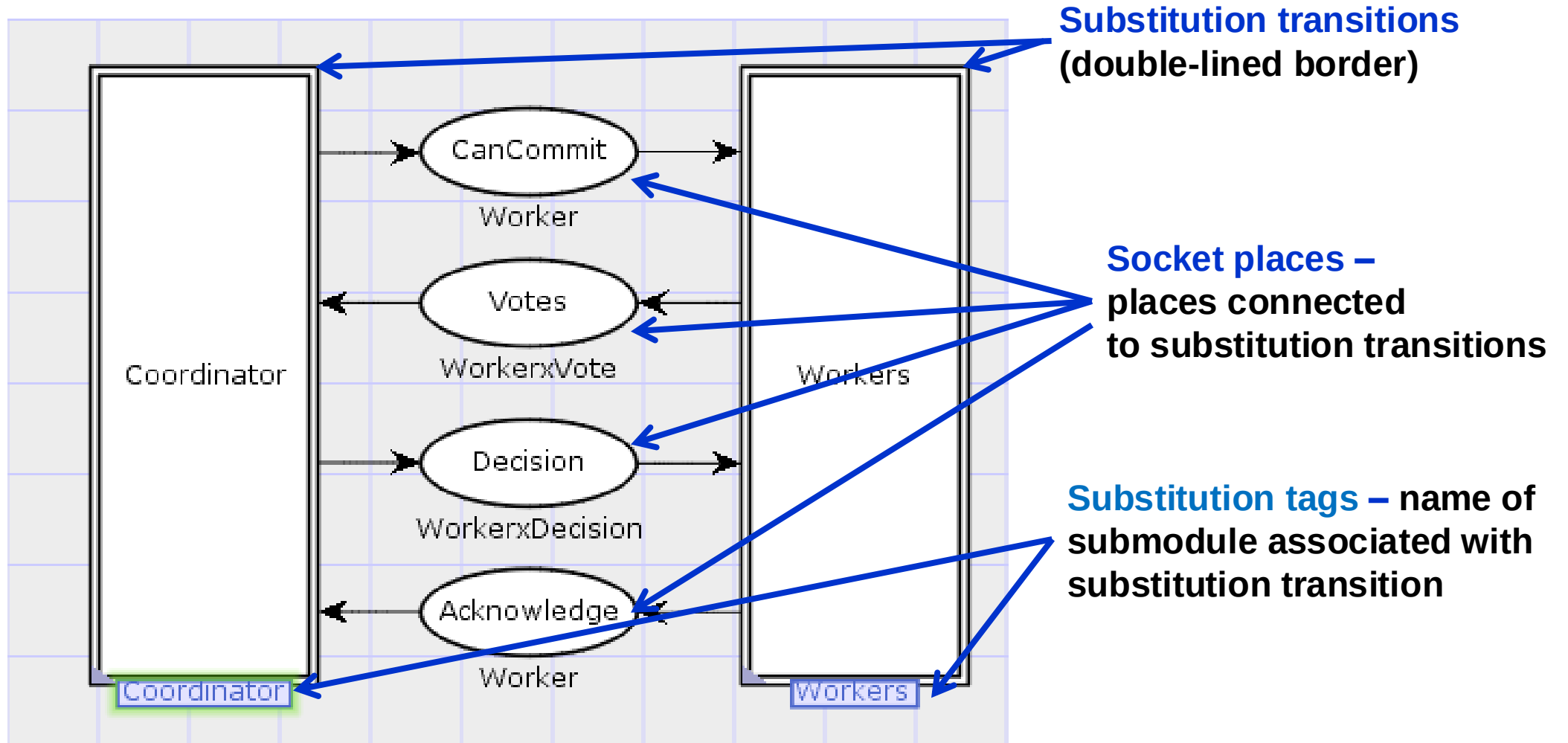
- **Important to be able to split a large CPN models into a set of modules with interfaces**
 - To support construction of large CPN models
 - To support reuse of modules and maintainability
 - To support abstraction and management of details
- **Key concepts of hierarchical CPN modules**
 - A **module** exchange tokens with its environment using input/output **port places**
 - **Substitution transitions** have associated **submodules**
 - **Port-socket relation** associates socket places of substitution transitions with the port places in the associated submodule

Hierarchical modules

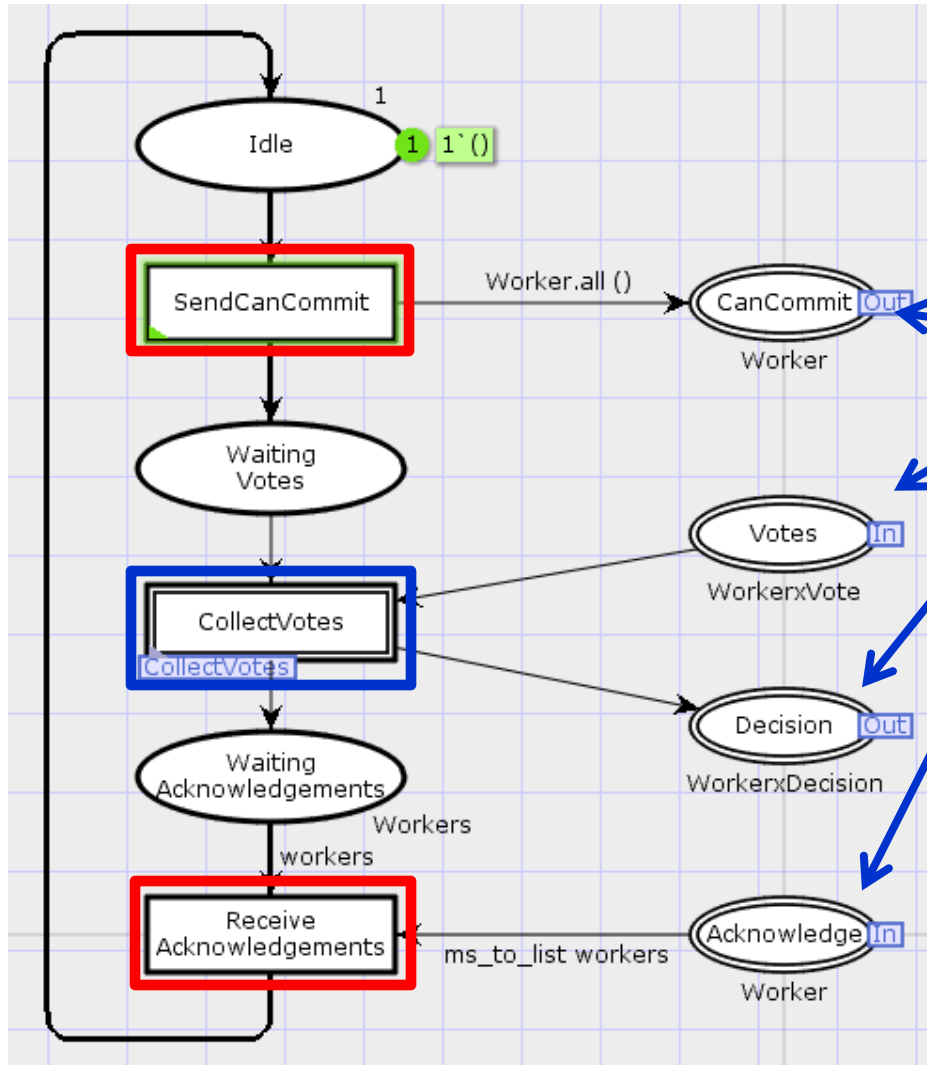
- Model is comprised of a collection of **modules** that are hierarchically organised into levels
- **Example:** the two-phase commit protocol



Top-level: Protocol module



Coordinator module



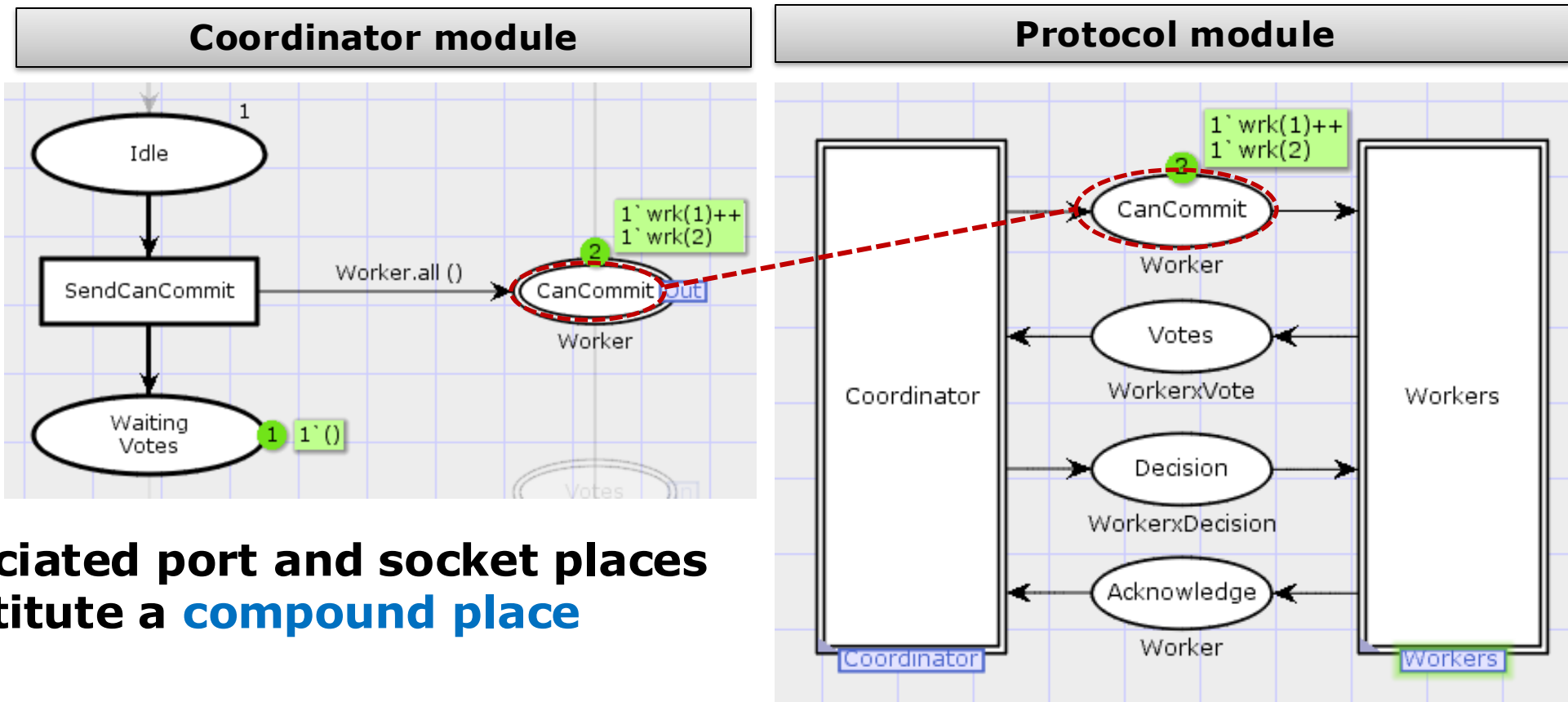
Port place - used for exchanging tokens with the upper-level module (IN,OUT,IN/OUT)

SendCanCommit and **ReceiveAcknowledgement** are ordinary transitions

CollectVotes is a substitution transition

Port-socket place relation

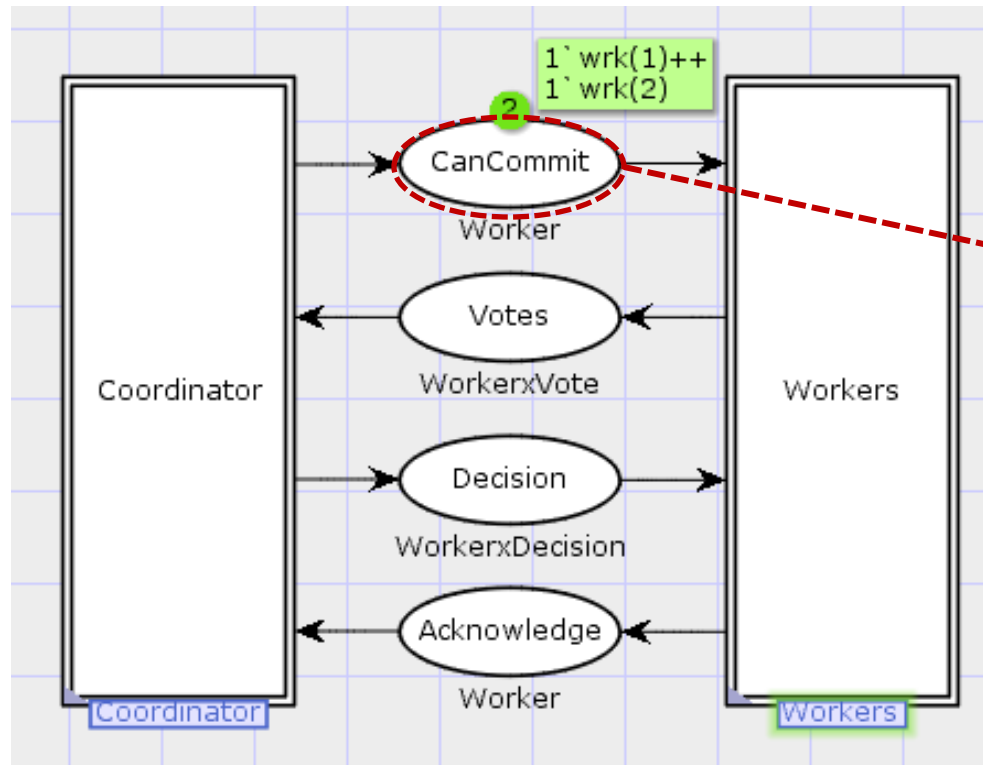
- Tokens added (removed) on a port place are added (removed) on the **associated socket place**



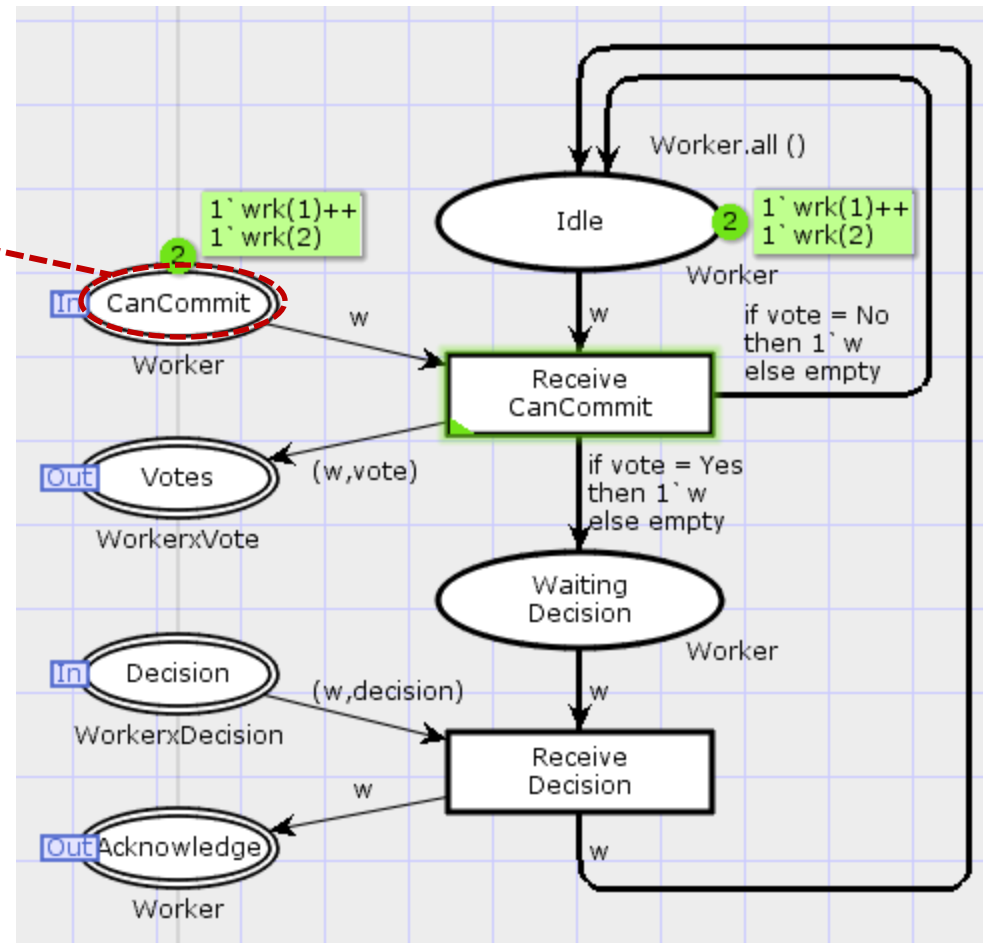
- Associated port and socket places constitute a **compound place**

Workers module

Protocol module



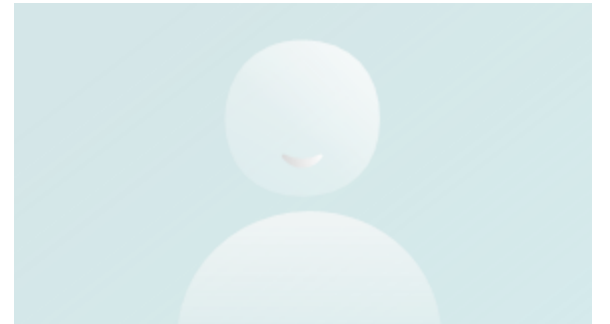
Workers module



CPN Tools – hierarchical modules

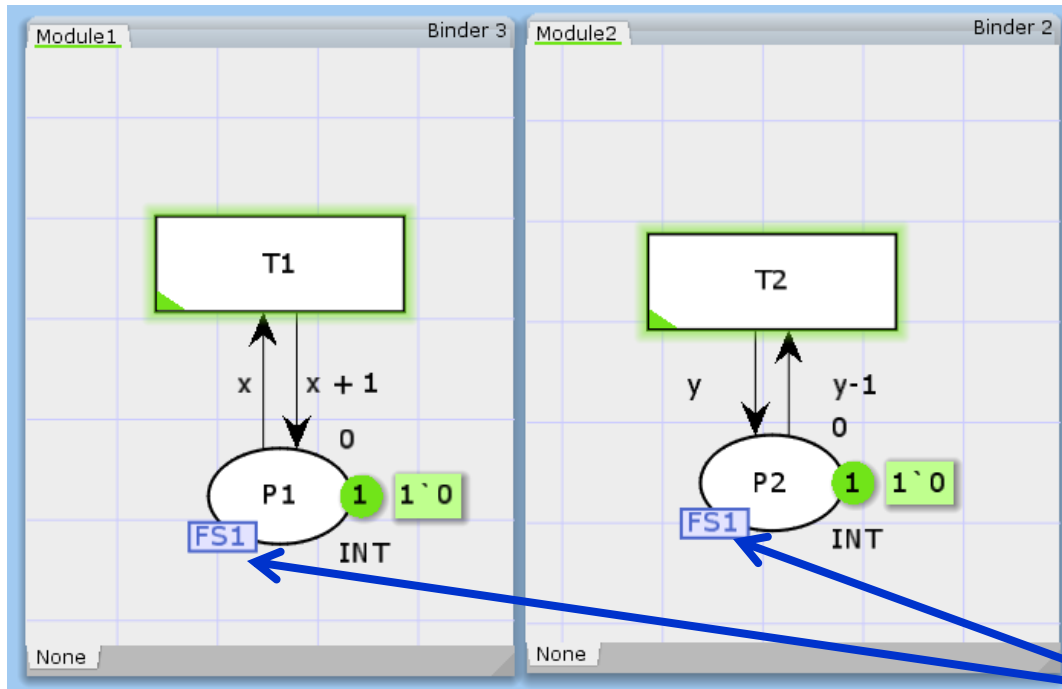
`hierarhical-modules-demo.cpn`

- **Hierarchical CPN models**
 - Navigating hierarchical models
 - Simulation of hierarchical models
 - Editing of modules: top-down and bottom-up development



Place Fusion Sets

- Group of places to be treated as one compound (global) place



Any change in the marking of P1 will be reflected on P2 (and vice versa)

Similar to global variables
- should be used with care

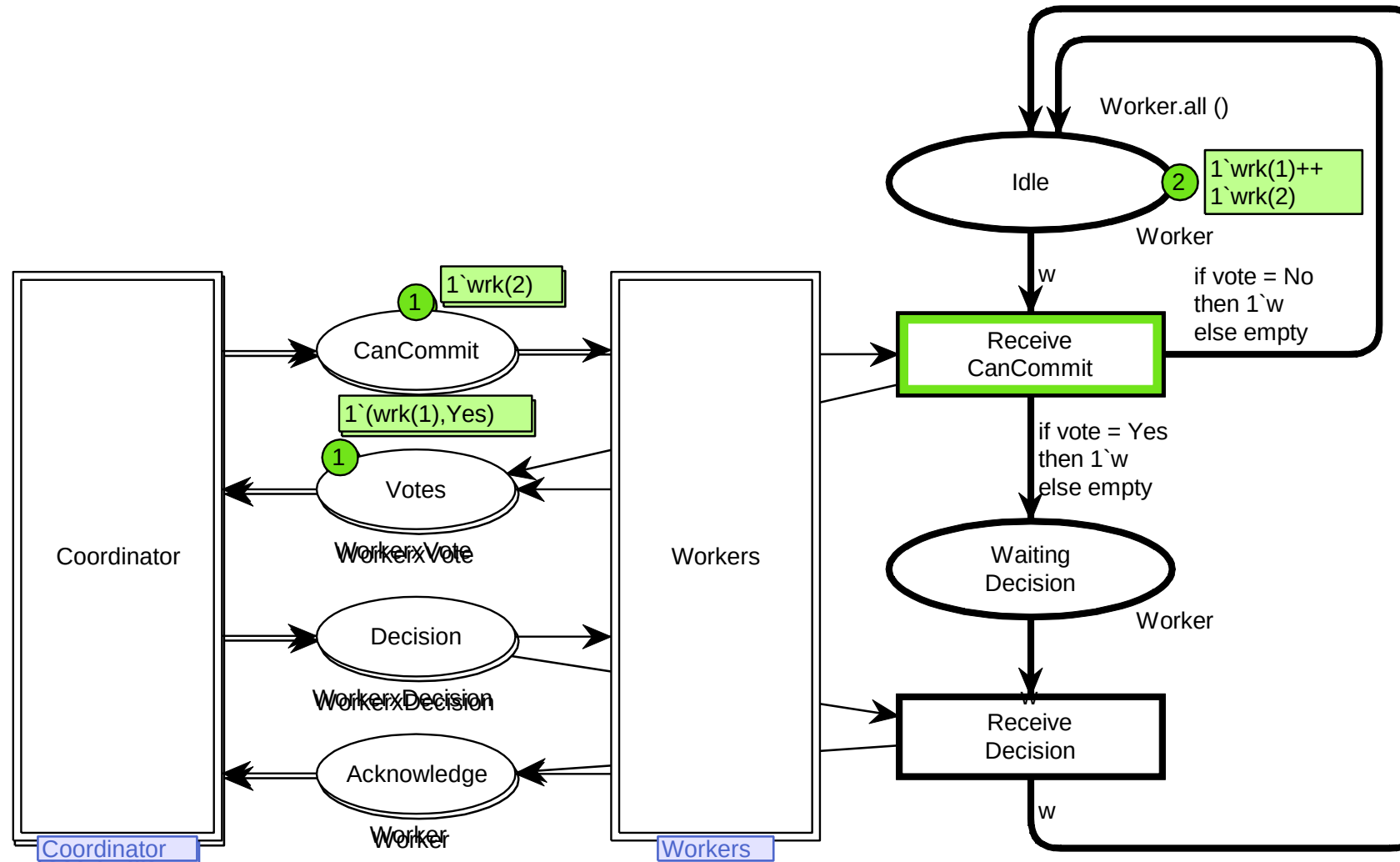
P1 and P2 are fusion places belonging to fusion set FS1

Unfolding Hierarchical Coloured Petri Nets to Place/Transition Nets

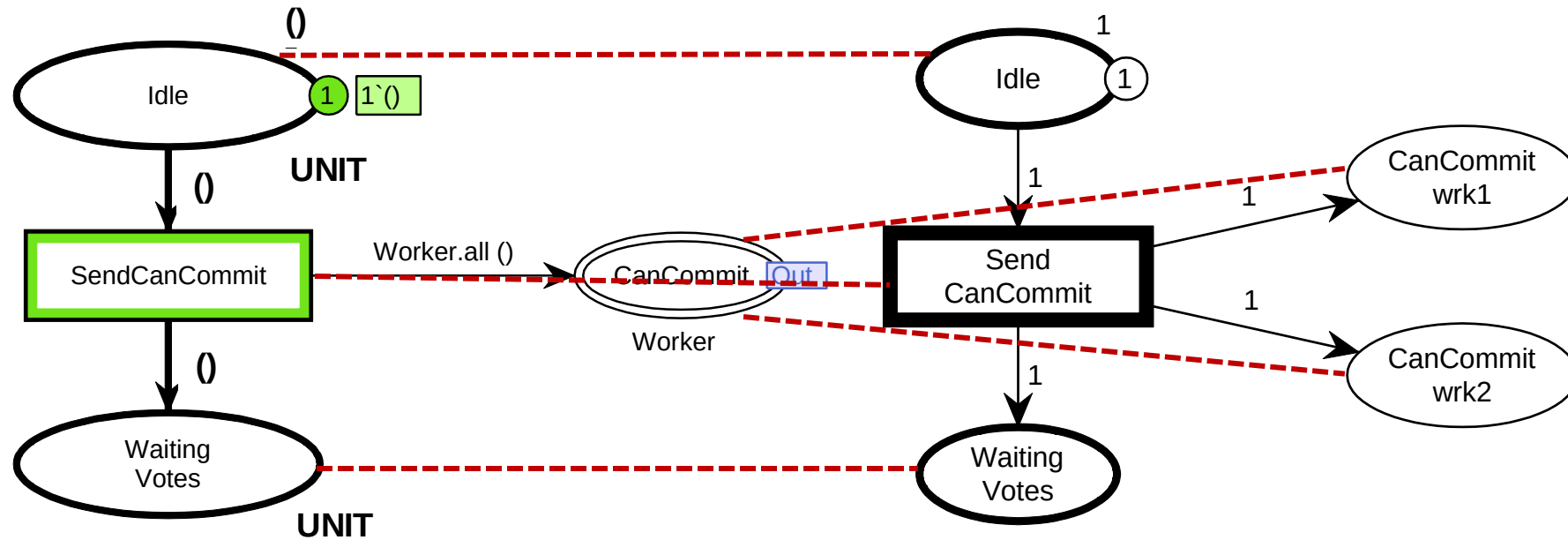
Unfolding Coloured Petri Nets

- **A hierarchical CPN model can be unfolded to a non-hierarchical Coloured Petri Net**
 - Recursively replace each substitution transition with its associated submodule
 - Associated port and socket places are merged into a single place
- **A non-hierarchical Coloured Petri Net can be unfolded into a Place/Transition Net (PTN)**
 - Replace each CPN place with one PTN place for each colour in the colour set of the CPN place
 - Replace each CPN transition with one PTN transition for each possible binding of the CPN transition

Unfolding hierarchical CPNs



Unfolding CPN places



Unfolding CPN transitions

